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Impact of Inter-terrace Conservation Measures on Rainwater Conservation, Runoff, Soil loss and Sorghum Yield on Vertisols under Dryland conditions of Semi Arid Tropics of South India

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ABSTRACT: Field experiments were conducted at the Research Farm of the Central Soil and Water Conservation Research Centre, Bellary, Karnataka during 2000-1 to 2002-03 in *rabi* season to study the impact of inter-terrace conservation measures on rain water conservation, runoff, soil loss, sorghum yield and soil properties. Decreased runoff and soil loss was observed in compartmental *bunding* + surface mulch treatment followed by graded border strips. Higher runoff and soil loss was recorded in control (up and down cultivation). Compartmental *bunding* + surface mulch and graded border strips increased the sorghum grain yield by 51.8 and 35.9% respectively, owing to reduction in runoff, soil loss and increased availability of moisture and nutrients, viz. N, P and K. Compartmental *bunding* + surface mulch and graded border strips recorded higher straw yield of 3043 kg ha^{-1} and 2868 kg ha^{-1} respectively. Soil moisture was conserved better in the top 60 cm in compartmental *bunding* + surface mulch. Higher moisture conserved with adoption of compartmental *bunding* + surface mulch resulted in higher values of yield components as compared to other treatments. Adoption of inter-terrace conservation measure, compartmental *bunding* + surface mulch reduced runoff and soil loss and improved the rainwater conservation, available nutrients, crop growth and yield.

Key words: Compartmental *bunding*, mulch, moisture, nutrients, runoff, soil loss, sorghum.

Soil and water are the two important resources on which the well being of human race as a whole depends. It is therefore, essential to conserve and use judiciously the above resources for higher yields. Dry lands are far more prone to erosion since they are devoid of vegetative cover for longer periods of time. Faulty land and crop management practices results in loss of nearly 5334 million tones (16.4 t ha^{-1}) of soil and 5.37 to 8.4 million tones of nutrients through erosion every year in India causing soil degradation (Dhruvanarayana and Ram Babu, 1983). Soil degradation can be reduced by reducing erosion and conserving rainwater through moisture conservation practices. Adoption of various *in-situ* moisture conservation measures at inter-terrace level will improve the soil moisture condition and nutrient status thereby stabilizing/sustaining crop yields in long run, especially during drought years. The soils of the Bellary region are clayey, with poor structure, low in infiltration rate, highly erodible, alkaline in reaction and cracking heavily on drying. Due to poor physical conditions, these soils are problematic and can be cultivated only in narrow range of soil moisture content. The rainfall is low (average annual rainfall is 508 mm), erratic and unevenly distributed. Due to low intake rate of soils, a maximum portion of the rainfall goes as runoff (15 to 20%) of the annual rainfall. The moisture is the limiting factor during crop growth resulting in crop failure. Frequent droughts and crop failure have become endemic in the region affecting the socio-economic conditions of the people. Hence, it is essential to conserve the excessive runoff at inter-terrace level by adopting suitable inter terrace conservation measures.

Rabi sorghum is an important dry land crop grown in Deccan plateau and it is the major source of food and

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fodder and as such it greatly influences the economic well being of people in the region. The production of the crop in general is low due to non-availability of moisture in the soil, improper management of rain water, adverse soil physical characters followed by poor nutrient status in the soil and subsistence farming practices which are adversely affecting economy in the region leading to unsustainability. In general, the slope in black soil region of semi-arid tropic of India varies from 1-3 per cent for cultivable lands and these soils are highly erodible. Irrespective of slope, mechanical measures are followed to conserve rainwater and reduce nutrient losses and degradation of land (Rao, 1997). In spite of adopting the mechanical measures, nearly 10 to 15% of rainwater is lost as runoff, which is very crucial to the success of post rainy season crops. It is therefore, essential to reduce runoff to a minimum by way of increasing opportunity time for the rainwater to soak into the profile either by adopting in-situ moisture conservation practices at inter-terrace level. Keeping in view a study was initiated to assess the impact of inter-terrace conservation measures on rainwater conservation, runoff, soil loss, soil properties and crop yields of *rabi* sorghum.

Materials and Methods

Field experiments were conducted at the Research Farm of the Central Soil and Water Conservation Research Centre, Bellary, Karnataka State, India during 2000 to 2003. The soil of the experimental site belongs to the Bellary series and is classified as fine montmorillonitic, gypsiferrous, isohyperthermic, typic pellusterts. The experiment was conducted on deep black soil and the clay content increased with soil depth from 46% at the surface to 57% at 60 to 95 cm. The soil pH ranged from 8.5 to 8.8 with available nutrient contents of N: 154 kg ha^{-1}, P: 12 kg ha^{-1} and K: 550 kg ha^{-1}. The four treatments were: T1-Up and down cultivation (Control), T2-Contour cultivation, T3-Formation of compartmental bunds (6m x 6m) before sowing of *rabi* crop including application of surface mulch (sorghum straw) and T4-Formation of border strips of 10m wide with no cross slopes having longitudinal gradient of 0.1 to 0.2%. The experiment was laid out in a completely randomized block design with five replications on 1.5% slope land. The individual plot size was 32.0 m x 13.0 m. Compartmental *bund*s and graded border strips were formed after preliminary tillage operations are completed during monsoon *i.e.* June to July to conserve rainwater *in situ*. Sorghum straw was applied as surface mulch in compartmental *bund*s. In contour cultivation treatment, preliminary tillage, sowing and interculturing operations were followed across the slope along the contour. Sorghum (*Sorghum bicolor* (L.) Moench. cv. SPV-86) was sown at 60 X 15 cm spacing with the onset of the northeast monsoon in the month of September 2000, 2001, and 2002 and were harvested on different dates in the following years depending upon maturity. Soil moisture was measured gravimetrically in the top 60cm soil depth at 30 days intervals from sowing to harvest. Grain yield and yield components of sorghum were recorded at harvest. The runoff and soil loss was collected every year with 60cm 'H' flume. Soil samples (0-15 and 15-30cm depths) were collected from all the plots after the crop harvest at the end of 3 years. These samples were analyzed for available nitrogen content by alkaline permanganate oxidation (Subbaiah and Asija, 1956), available phosphorus content by Olsen's method (Jackson, 1967) and available potassium content by extracting with neutral normal ammonium acetate, photometrically (Muhr, 1965)

Results and Discussion

Runoff and soil loss

There were 3 runoff events in 2000-01, 9 runoff events in 2001-2002 and 8 in 2002-2003. Total runoff causing rainfall was 432.6, 566.6, and 550.9 mm during 2000-01, 2001-2002, and 2002-2003, respectively. Lower runoff (33.9 mm) and soil loss (1191 kg ha^{-1}) occurred in compartmental *bunding* + surface mulch (**Table 1**), followed by graded border strips. Contour cultivation, graded border strips, and compartmental *bund*ing + surface mulch have reduced runoff by 33.7%, 44.7.5%, and 58.1% over up and down cultivation (control) respectively. Contour cultivation, graded border strips, and compartmental *bund*ing + surface mulch have reduced soil loss by 24.1%, 36.7% and 52.1%, over up and down cultivation (control) respectively. The higher runoff (62.2 mm) and soil loss (2845 kg ha^{-1}) was observed in the control plot (up and down cultivation). Similar trend was observed during all the years of study. Rama Mohan Rao *et al*. (2000) also observed highest

Table 1 : Runoff and soil loss as influenced by inter-terrace conservation measures.

Treatments	Runoff (mm)				Soil loss (kg ha^{-1})			
	2000-01	2001-02	2002-03	Mean	2000-01	2001-02	2002-03	Mean
T1-Up & down cultivation	57.98	62.99	65.60	62.19	2394	3500	2640	2845
T2- Contour cultivation	47.27	51.93	32.71	43.97	1280	2500	1875	1885
T3- Compartmental *bund*ing+ Surface mulch	38.96	43.73	19.09	33.93	704	1618	1250	1191
T4- Graded border strips (10 m wide)	46.48	48.79	28.78	41.35	1060	2006	1650	1572
Runoff causing rainfall (mm)	150.7	282.4	225.7					

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runoff and soil loss of 55.5 mm and 1644 kg ha^{-1} in up and down cultivation and lowest (23.7mm and 224 kg ha^{-1}) runoff and soil loss under vertical mulch + surface mulch in the Vertisols of Bellary. Katama Reddi and Padmalatha (1993) reported that it could be possible to reduce soil and water losses by adopting in-situ conservation practices like dead furrows at 3.6m interval across the slope (25 to 30 days after sowing) with the arrival of rains coupled with compartmental bunding with 20m length and 10m width before germination of crop. Adhikari *et al*. (2002) observed that sorghum stover mulch reduced runoff and soil loss considerably and concluded that simple application of mulch material on the surface up to 2% slope of field, can act as good inter terrace soil and water conservation treatment in the deep black soils of Bellary, apart from improving in-situ moisture condition to increase crop production.

Soil moisture and yield

Compartmental *bunding* + surface mulch treatment recorded higher yield in all the three seasons (200-01, 2001-02 and 2002-03). Sorghum grain and straw yields were higher during 2001-02 as compared to 2000-01 and 2002-03 was attributed to higher rainfall (566.6mm) received during 2001-02 with higher antecedent (189.0mm) and crop season rainfall (148.5mm). Grain and straw yields of *rabi* sorghum increased significantly by 35.9% (1212 kg ha^{-1}) and 44.6% (2868 kg ha^{-1}) with formation of graded border strips and further improved to 51.8 and 53.5% (1354 and 3043 kg ha^{-1}) with compartmental *bund*ing + surface mulch compared to up and down cultivation (892 and 1983 kg ha^{-1}) respectively (Table 2). Higher yields in the above treatments were due to higher moisture conserved with adoption of *in situ* moisture conservation practice at inter terrace level and higher dry matter production and its translocation to reproductive parts resulting in higher values of yield components. The yield components also followed similar trend as that of grain yield. Compartmental *bund*ing + surface mulch and graded border strips recorded higher panicle length, panicle weight, panicle diameter and 1000 seed weight than other treatments (Table-3). Higher yields in the plots laid out with inter terrace conservation measures viz. compartmental *bund*ing + surface mulch and graded border strips was due to conservation of rain water *in-situ* which has further led to higher moisture availability in the profile throughout the crop season. Higher soil moisture in top 60 cm soil profile was observed in the plots with compartmental *bund*ing + surface mulch during all the stages of crop growth in all the years of study (Table-4). The treatment compartmental *bund*ing + surface mulch recorded higher soil moisture in top 60 cm soil profile during all the three seasons with the average soil moisture of 19.4 cm at sowing, 15.5 cm at boot leaf stage, 12.9 cm at grain formation stage and 10.6 cm at harvest. Up and down cultivation (Control) recorded lower soil moisture in top 60 cm soil profile during all the three seasons with the average soil moisture of 17.0 cm at sowing, 11.9 cm at boot leaf stage, and 9.7 cm at grain formation stage and 7.6 cm at harvest. These results indicated that the treatment with compartmental *bund*ing + surface mulch proved

Table 2 : Grain and straw yield (kg ha^{-1}) of *rabi* sorghum as influenced by inter-terrace conservation measures.

Treatments	Grain yield (kg ha^{-1})				Straw yield (kg ha^{-1})			
	2000-01	2001-02	2002-03	Mean	2000-01	2001-02	2002-03	Mean
T_1	400	1634	542	859	1017	3147	1786	1983
T_2	440	1737	647	941	1567	3250	1820	2212
T_3	938	1999	1158	1365	2431	3858	2840	3043
T_4	760	2137	992	1296	2264	3600	2740	2868
SEm ±	5.01	9.71	4.31	6.40	11.24	9.40	10.46	10.37
CD (P=0.05)	15.43	29.91	13.27	19.54	34.64	28.96	32.22	31.94
Total rainfall (mm)	432.6	566.6	550.9					
Antecedent rainfall (mm)	125.7	189.0	142.7					

Table 3 : Yield components of *rabi* sorghum (Pooled 2000-01 to 2002-03) as influenced by inter-terrace conservation measures.

Treatments	Panicle weight (g)	Panicle length (cm)	Panicle diameter (cm)	1000 seed weight (g)
T_1	30.8	12.7	11.6	22.5
T_2	36.7	13.9	13.2	26.6
T_3	48.1	15.4	15.8	28.4
T_4	40.9	14.2	14.1	27.5
SEm ±	2.25	0.37	0.74	1.08
CD (P=0.05)	6.93	1.14	2.27	3.32

Table 4 : Total Soil moisture (cm) in 60 cm soil profile at different stages of plant growth as influenced by inter-terrace conservation measures.

Treat-mentss	At sowing			Boot leaf stage			Grain formation stage			At harvest		
	2000-01	2001-02	2002-03	2000-01	2001-02	2002-03	2000-01	2001-02	2002-03	2000-01	2001-02	2002-03
T_1	16.4	17.6	16.9	11.5	12.3	11.9	9.0	10.8	9.4	6.7	8.6	7.5
T_2	16.9	18.5	18.1	12.7	13.4	13.0	10.1	11.7	11.0	7.6	9.5	8.6
T_3	18.9	19.9	19.5	14.9	16.1	15.6	12.2	14.1	12.5	9.5	11.9	10.5
T_4	18.0	19.4	18.9	13.8	14.9	14.2	11.2	13.3	11.7	8.5	11.0	9.0

beneficial in conserving more rainwater as compared to the rest of the treatments with greater sorghum grain and straw yields. Similar results were reported by More *et al*. (1996) by assessing the use of compartmental *bunding* (6 X 6 m) as a moisture conservation measure for increased productivity of winter sorghum on farmers' fields at Solapur, Maharashtra and concluded that compartmental *bunding* increased mean sorghum yield from 0.69 to 0.96 t ha^{-1} and mean fodder yield from 1.2 to 1.79 t ha^{-1}. Selvaraju *et al*. (1999) reported that compartmental *bunding* stored 22% more soil moisture and increased the yield of sorghum + pigeon pea intercropping than did flat bed in a low rainfall year. Ravi and Pandurangaiah (2001) observed an additional yield of 5.8 and 3.7q per ha of grain and straw, respectively by following graded border strips over the farmers practice in Rajanukunte watershed, Bangalore. Higher yields with adoption of *in-situ* moisture conservation practices viz. compartmental *bunding* and ridges and furrows were also reported by Velayudham *et al.* (1997) and Patil and Sheelvantar (2001).

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Table 5 : Available nutrients (kg ha^{-1}) at different depths at the end of three years as influenced by inter-terrace conservation measures.

Treatments	Available Nutrients (kg ha^{-1})					
	Nitrogen		Phosphorus		Potassium	
	0-15 cm	15-30 cm	0-15 cm	15-30 cm	0-15 cm	15-30 cm
T_1	165.6	155.7	17.8	17.2	558	540
T_2	173.4	158.5	20.4	19.3	565	542
T_3	184.5	165.2	21.5	20.2	583	549
T_4	177.2	161.9	20.9	19.8	572	546
SEm ±	2.76	3.20	0.86	1.06	2.95	3.05
CD (P=0.05)	8.51	NS	2.64	NS	9.09	NS

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Nutrients

Crop yields in drylands not only depend upon the moisture but also on the availability of nutrients in the profile. Contour cultivation, graded border strips, and compartmental *bund*ing + surface mulch have increased the available nitrogen by 7.8, 11.6 and 18.9 kg ha^{-1} and available phosphorus by 2.6, 3.1 and 3.7 kg ha^{-1} and available potassium by 7, 14 and 25 kg ha^{-1}over up and down cultivation (control) respectively (Table 5). The cause for higher nutrient availability in the plots where inter-terrace conservation practices were adopted was due to less loss of nutrients through runoff and soil loss and higher rate of mineralisation. Jayaram *et al.* (1982) quoted that contour cultivation alone reduced the soil loss from 12.5 to 2.0 tones ha^{-1} over up and down cultivation thus reducing the loses of nutrients of total nitrogen, available phosphorus and potassium and exchangeable calcium and magnesium in the Vertisols of Bellary. Similarly in the Vertisols of Akola, contour cultivation increased the residual nitrogen by 39 kg ha^{-1}, available phosphorus by 4.6 39 kg ha^{-1} and available potassium by 42 kg ha^{-1} respectively over cultivation along the slope (Sagare *et al.* 1992). In the present investigation, compartmental *bund*ing + surface mulch and graded border strips signicantly improved the available N, P and K over up and down cultivation in top soil (0-15cm) after harvest of sorghum crop at the end of three years. However, subsoil (15-30 cm) nutrient availability was lower than top soil probably due to lower organic matter content, whereas higher availability of nutrients in top soil was due to application of fertilizers and higher rate of mineralisation. The above results are in conformity with findings of Patil and Sheelvantar (2001).

Experimental results in Vertisols of Bellary, Karnataka state, India indicated that compartmental *bund*ing + surface mulch, reduced runoff and soil loss and improved the rainwater conservation, available nutrients, crop growth and yield of *rabi* sorghum in dry land conditions.

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